

# ChatGPT - DSA swe

Give me visualization of step by step deletion process in trie

Sure! Let's visualize the **step-by-step deletion process in a Trie** with a clear example. We'll walk through inserting some words and then deleting one of them.



Let's start with a Trie containing these words:

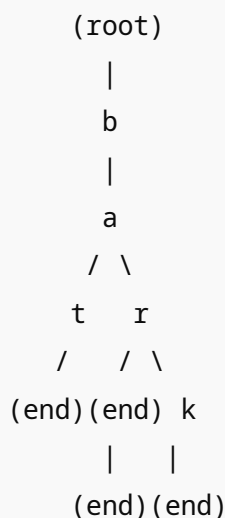
- "bat"
- "bad"
- "bark"
- "bar"



## Step 1: Insert the words into Trie

We insert character by character. Here's how the trie looks:

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- "bat" ends at **t**
- "bad" shares b-a with "bat", then diverges to **d**

- "bar" shares b-a and ends at **r**
- "bark" extends "bar" with **k**

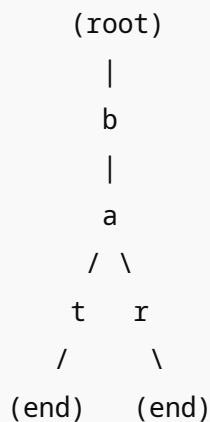
## ❌ Step 2: Delete "bark"

### Step-by-step logic:

- Go character by character: **b** → **a** → **r** → **k**
- Start from the bottom and go up:
  - Is 'k' a leaf node? ☒ Yes → delete node **k**
  - Is 'r' now a leaf and not end of any other word? ☒ No, because "bar" ends at **r** → stop deletion here.

### Updated Trie:

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- ✓ "bark" is deleted
- ✓ "bar" is still valid

## 🧠 Final Notes:

- Deletion in Trie is **recursive and bottom-up**.
- A node is deleted **only if**:

- It's not marked as end of another word
- It has no children (leaf)



## Want a graphic diagram?

I can generate a **step-by-step image visualization** if you'd like! Just say the word and I'll draw the before and after diagrams for each step.



stop

No problem! Let me know if you'd like to continue with the visualization later or need anything else explained or illustrated.

give me more different typesa of example

Absolutely! Here are **different types of Trie deletion examples**, each covering a unique case. These will help you understand all the edge cases in the deletion process:

### ✅ Example 1: Delete a word that is a prefix of another word

Inserted words: "car", "care"

Delete: "car"

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Initial Trie:

```
(root)
  |
  c
  |
  a
  |
  r  --(end for "car")
  |
  e  --(end for "care")
```

After Deleting "car":

- We **do not delete** `r`, because it's needed for "care"
- Just **remove the end-of-word mark** on `r`

Final Trie remains intact:

"care" works, "car" is no longer recognized

### ✅ Example 2: Delete a word that shares prefix but no branch

Inserted words: "car", "cab"

Delete: "car"

vbnet

Initial Trie:

```
(root)
  |
  c
  |
  a
 / \
r   b
(end)(end)
```

After Deleting "car":

- Remove node `r` since it's a leaf and only used by "car"

Final Trie:

"cab" remains, "car" is gone

### ✓ Example 3: Delete a word with no shared prefix

Inserted words: "dog", "cat"

Delete: "cat"

sql

Initial Trie:

```
(root)
  |
  ├── c → a → t (end)
  └── d → o → g (end)
```

After Deleting "cat":

- Entire branch "c-a-t" is deleted

Final Trie:

Only "dog" remains

## ✓ Example 4: Delete a word that is a superstring of another

Inserted words: "do", "dog"

Delete: "dog"

sql

Initial Trie:

```
(root)
  |
  d
  |
  o --(end for "do")
  |
  g --(end for "dog")
```

After Deleting "dog":

- Delete node `g` (leaf)
- Keep node `o` because it ends "do"

Final Trie:

"do" remains, "dog" is gone

## ✓ Example 5: Try to delete a word that doesn't exist

Inserted words: "bat", "bar"

Try to delete: "bag"

vbnet

Initial Trie:

```
(root)
  |
  b
  |
  a
  / \
```

```
    t   r  
    (end)(end)
```

"bag" path doesn't *exist*.

Result:

- Nothing is deleted
- Trie remains unchanged

Would you like visual diagrams for any of these examples as well? I can draw them like before!